In JEE Advanced 2016, FIITJEE Students bag 36 in Top 100 AIR, 75 in Top 200 AIR, 183 in Top 500 AIR. 3541 Students from Long Term Classroom/ Integrated School Program & 4423 Students from All Programs have qualified in JEE Advanced, 2016

FIITJEE

JEE (Advanced), 2017

ALL INDIA TEST SERIES

FULL TEST – II

Paper 1

Time Allotted: 3 Hours **Maximum Marks: 234**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

Caution: Question Paper CODE as given above MUST be correctly marked in the answer OMR sheet before attempting the paper. Wrong CODE or no CODE will give wrong results.

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part is further divided into two sections: **Section-A & Section-C**
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

(i) **Section-A (01 – 08)** contains 8 multiple choice questions which have only one correct answer. Each question carries **+3 marks** for correct answer and **– 1 mark** for wrong answer.

Section-A (09 – 12) contains 4 multiple choice questions which have more than one correct answer. Each question carries **+4 marks** for correct answer. There is no negative marking.

Section-A (13 to 18) contains 3 paragraphs with each having 2 questions. Each question carries **+3 marks** for correct answer and **– 1 mark** for wrong answer.

(ii) **Section-C (01 – 05)** contains 5 Numerical based questions with answers as numerical value and each question carries **+4 marks** for correct answer. There is no negative marking.

Name of the Candidate	
Enrolment No.	

Useful Data

PHYSICS

Acceleration due to gravity	$g = 10\ m/s^2$
Planck constant	$h = 6.6 \times 10^{-34}$ J-s
Charge of electron	$e = 1.6 \times 10^{-19}$ C
Mass of electron	$m_e = 9.1 \times 10^{-31}$ kg
Permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$
Density of water	$\rho_{water} = 10^3\ kg/m^3$
Atmospheric pressure	$P_a = 10^5\ N/m^2$
Gas constant	$R = 8.314\ J\ K^{-1}\ mol^{-1}$

CHEMISTRY

Gas Constant	R	=	$8.314\ J\ K^{-1}\ mol^{-1}$
		=	0.0821 Lit atm $K^{-1}\ mol^{-1}$
		=	$1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$
Avogadro's Number	N_a	=	6.023×10^{23}
Planck's constant	h	=	6.625×10^{-34} J·s
		=	6.625×10^{-27} erg·s
1 Faraday		=	96500 coulomb
1 calorie		=	4.2 joule
1 amu		=	1.66×10^{-27} kg
1 eV		=	1.6×10^{-19} J

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

Physics

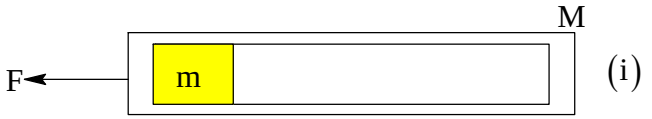
PART – I

SECTION – A

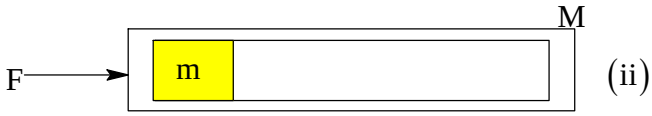
Straight Objective Type

This section contains **08 multiple choice questions** numbered 1 to 8. Each question has 4 choices (A), (B), (C) and (D), out of which only **ONE** is correct

1. A block of mass m is placed in contact with one end of a smooth tube of mass M (see figure). A horizontal force F acts on the tube in each case (i) and (ii). Then mark **incorrect** option:



(i)



(ii)

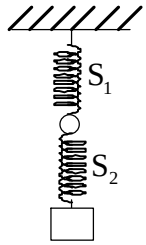
(A) $a_m = 0$ and $a_M = \frac{F}{M}$ in (i)

(B) $a_m = a_M = \frac{F}{M+m}$ in (i)

(C) $a_m = a_M = \frac{F}{M+m}$ in (ii)

(D) Force on m is $\frac{mF}{M+m}$ in (ii)

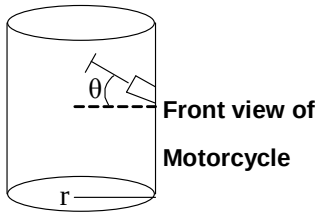
2. Two springs, S_1 and S_2, have negligible masses and the spring constant of S_1 is $1/3$ that of S_2. When a block is being hanged from the springs as shown above and the springs slowly come to equilibrium again, the ratio of the potential energy stored in S_1 to the S_2 is:



(A) 1/9
(B) 1/3
(C) 1
(D) 3

Space for rough work

3. A trick cyclist rides his bike around a well of death in the form of a vertical cylinder (see figure). The maximum frictional force parallel to the surface of the cylinder is equal to a fraction μ of the normal force exerted by the wall. At what minimum constant speed must the cyclist go to avoid slipping down? (Assume that size of motorcycle is very small as compared to cylinder.



(A) $\left[\frac{2gr\cos\theta}{\mu}\right]^{1/2}$ (B) $\left[\frac{gr\sin\theta}{2\mu}\right]^{1/2}$

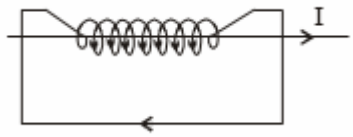
(C) $\left[\frac{gr}{\mu}\right]^{1/2}$ (D) $\left[\frac{2gr}{\mu}\right]^{1/2}$

4. A thermometer consists of a quantity of alcohol in a glass tube. When the thermometer is in thermal contact with soft ice cream in a box the length of the alcohol column is 5 cm and thermal equilibrium. When placed in a freezer the alcohol column remains 5 cm long in thermal equilibrium. When ice cream is put in the freezer.

(A) Ice cream melts (B) Ice cream freezes more

(C) Ice cream neither melts nor freezes more (D) Insufficient information.

5. In the diagram shown, a wire carries current I. What is the value of the $\oint \vec{B}.d\vec{s}$ (as in Ampere's law) on the helical loop shown in the figure? The integration in done in the sense shown. The loop has N turns and part of helical loop on which arrows are drawn is outside the plane of paper.



(A) $-\mu_0(NI)$ (B) $\mu_0(I)$

(C) $\mu_0(NI)$ (D) Zero

6. A projectile is thrown with velocity U = 20m/s ± 5% at an angle 60°. If the projectile falls back on the ground at the same level then which of following can not be a possible answer for range.

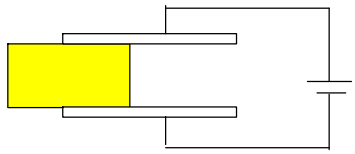
Consider $g = 10 m/s^2$.

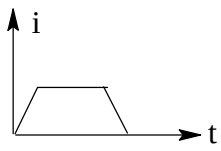
(A) 39.0 m (B) 37.5 m

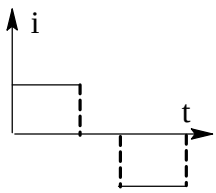
(C) 34.6 m (D) 32.0 m

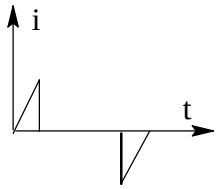
Space for rough work

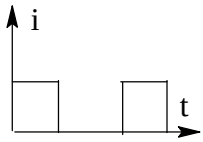
7. A dielectric slab of area A passes between the capacitor plates of area 2A with a constant speed v. The variation of current (i) through the circuit as function of time (t) can be qualitatively represented as



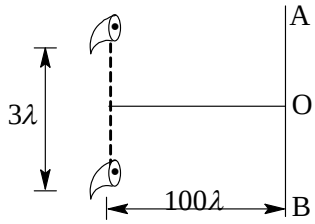
(A) 

(B) 

(C) 

(D) 

8. 2 loudspeakers are emitting sound waves of wavelength λ with an initial phase difference of $\frac{\pi}{2}$. At what minimum distance from O on line AB will one hear a maxima



(A) 25λ

(B) $\frac{100\lambda}{\sqrt{15}}$

(C) $\frac{25\lambda}{3}$

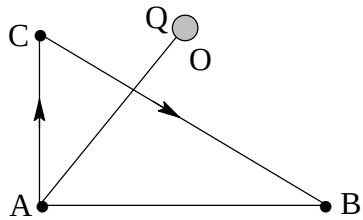
(D) 50λ

Space for rough work

Multiple Correct Choice Type

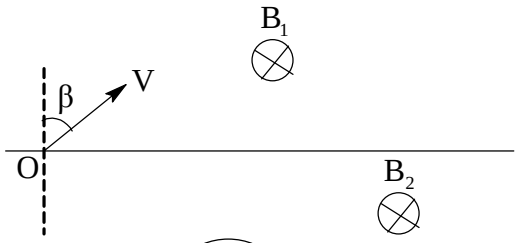
This section contains **04 multiple choice questions.** Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which only **ONE OR MORE THAN ONE** is/are correct

9. There is a fixed positive charge Q at O and A and B are points equidistant from O. A positive charge + q is taken slowly by an external agent from A to B along the line AC and then along the line CB.

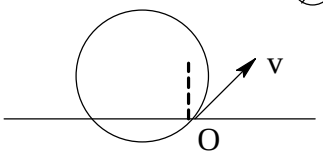


(A) The total work done on the charge is zero
(B) The work done by the electrostatic force from A to C is negative
(C) The work done by the electrostatic force from C to B is positive
(D) The work done by electrostatic force in taking the charge from A to B is dependent on the actual path followed.

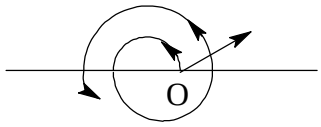
10. In the diagram shown, a particle of charge +Q and mass M is projected making an angle β with the vertical line. Which the possible path on which the charge will move. Above the dark line magnetic field is B_1 and below it is B_2. (Consider all possible cases for values of B_1 and B_2)



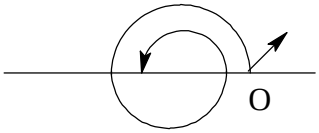
(A)



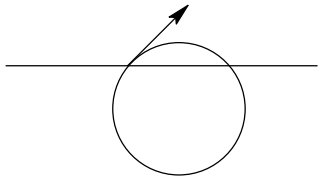
(B)



(C)



(D)



Space for rough work

11. Two particles of mass m1 and m2 are moving with velocity $\vec{v}_1$ and $\vec{v}_1$ respectively. Mark the correct statement about a perfectly inelastic collision between m_1 and m_2.

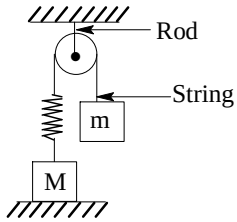
(A) Impulse by m_1 to m_2 is $\frac{m_1 m_2}{m_1 + m_2}(\vec{v}_1 - \vec{v}_2)$

(B) Momentum transferred by m_2 to m_1 is $\frac{m_1 m_2}{m_1 + m_2}(\vec{v}_2 - \vec{v}_1)$

(C) Impulse by m_1 on m_2 in CM frame is $\frac{m_1 m_2}{m_1 + m_2}(\vec{v}_2 - \vec{v}_1)$

(D) In CM frame both the particles come to rest after perfectly inelastic collision.

12. In figure, a block of mass m is released from rest when spring was in its natural length. The pulley also has mass m but it is frictionless. Suppose the value of m is such that finally it is just able to lift the block M up after releasing it.



(A) The weight of m required to just lift M is $\frac{M}{2}g$

(B) The tension in the rod, when m is in has zero acceleration $\frac{M}{2}g$

(C) The normal force acting on M when m has zero acceleration $\frac{M}{2}g$

(D) The tension in the string when displacement of m is maximum possible is Mg

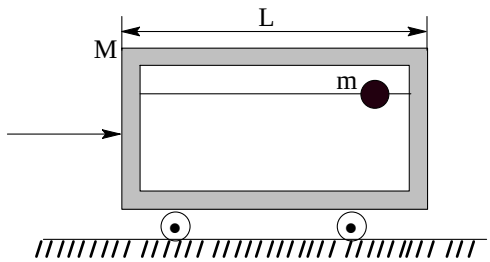
Space for rough work

Comprehension Type

This section contains **3 groups of questions.** Each group has 2 multiple choice question based on a paragraph. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which only **ONE** is correct

Paragraph for Questions 13 & 14

A horizontal frictionless rod is threaded through a bead of mass m. The length of the cart is L and the radius of the bead, r, is very small in comparison with L (r << L). Initially at (t = 0) the bead is at the right edge of the cart. The cart is struck and as a result, it moves with velocity υ_0. When the bead collides with the cart's walls, the collisions are always completely elastic. (Given: m > M).



13. What is the velocity of the cart after the first collision, in the center of mass frame?

(A) $\frac{-m v_0}{m+M}$ (B) $\frac{M v_0}{m+M}$

(C) $\frac{M-m}{M+m} v^0$ (D) $\frac{2M}{m+M} v^0$

14. The first collision takes place at time t1 and the second collision takes place at $t = 0$ till $t = t_2$ if $m > M_0$

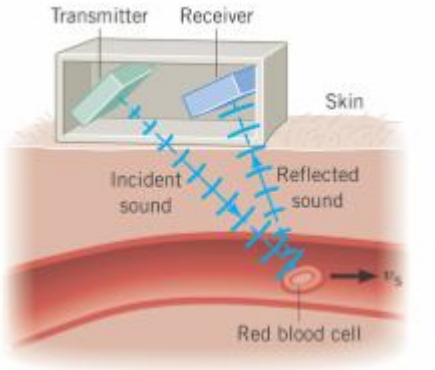
(A) $\frac{2ML}{m+M}$ (B) $\frac{ML}{2(m+M)}$

(C) $\frac{2M}{m+M} L$ (D) $\frac{ML}{2(m+M)}$

Space for rough work

Paragraph for Questions 15 & 16

The Doppler flow meter is a particularly interesting medical application of the Doppler effect. This device measures the speed of blood flow, using transmitting and receiving elements that are placed directly on the skin, as in Figure. The transmitter emits a continuous sound whose frequency is typically about 5 MHz. When the sound is reflected from the red blood cells, its frequency is changed in a kind of Doppler effect because the cells are moving with the same velocity as the blood. The receiving element detects the reflected sound, and an electronic counter measures its frequency, which is Doppler-shifted relative to the transmitter frequency. From the change in frequency the speed of the blood flow can be determined. Typically, the change in frequency is around 600 Hz for flow speeds of about 0.1 m/s.



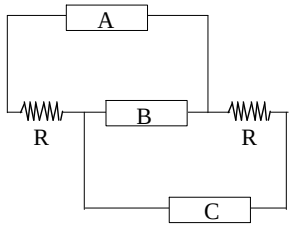
15. Assume that the red blood cell is directly moving away from the source and the receiver. What is the (approx) speed of the sound wave in the blood?
(A) 1700 m/s (B) 330 m/s
(C) 5000 m/s (D) 3000 m/s

16. An abnormal segment of the artery is narrowed down by an arteriosclerotic plaque to one-fourth the normal cross-sectional area. What will be the change in frequency due to reflection from the red blood cell in that region?
(A) 150 Hz (B) 300 Hz
(C) 600 Hz (D) 2400 Hz

Space for rough work

Paragraph for Questions 17 & 18

A circuit is shown below.



17. If A is an ideal ammeter, B an ideal Battery of voltage V, and C an ideal voltmeter, what will be the $\frac{\text{reading of C}}{\text{reading of A}}$?

(A) R (B) 2R

(C) $\frac{R}{2}$ (D) 0

18. If A is a capacitor, B is an ideal ammeter and C is an ideal battery of voltage V, what is the voltage across the capacitor ?

(A) V (B) $\frac{V}{2}$

(C) 2V (D) 0

Space for rough work

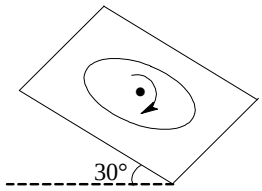
SECTION – C

(One Integer Value Correct Type)

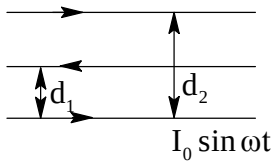
This section contains **5 questions**. Each question, when worked out will result in **one integer** from 0 to 9 (both inclusive).

1. In a slow reaction, heat is being evolved at a rate about 10mW in a liquid. If the heat were being generated by the decay of ^{32}P, a radioactive isotope of phosphorus that has half-life of 14 days and emits only beta-particles with a mean energy of 700KeV. The number of ^{32}P atoms in the liquid is $A\times 77\times 10^{15}$ find a:
[Take: ln 2 = 0.7]

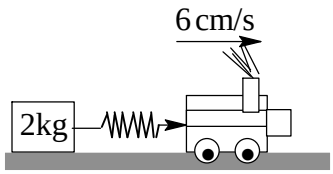
2. An old record player of 10 cm radius turns at 10 rad/s while mounted on a 30° incline as shown in the figure. A particle of mass m can be placed anywhere on the rotating record. If the least possible coefficient of friction μ that must exist for no slipping to occur is μ, find $2\sqrt{3}\mu$.



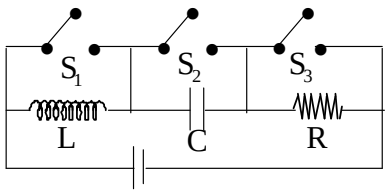
3. Two long straight wires of telephone circuit inside a house are coplanar with and run parallel to a third long straight conductor that carries a current $i = i_0 \sin \omega t$. The wires are at a distance of d_1 and d_2 from the conductor. The conductor induced a noise in the telephone circuit. The rms voltage per unit length of this induced emf is $x\times 5$, find x? Express your answer in μV. Take $d_2 = 2d_1, i_0 = \frac{2\sqrt{2}}{\ln 2}\text{A}\ \omega = 50\ \text{rad/s}$



4. Figure a shows a spring attached to a 2.0 kg block. The other end of the spring is pulled by a motorized toy train that moves forward at constant speed 6.0 cm/s. The spring constant is 50 N/m and the coefficient of static friction between the block and the surface is 0.60. The spring is at its natural length at t = 0s when the train starts to move. When does the block slip (express your answer in second)?



5. Consider the circuit shown in figure. With switch S_1 closed and the other two switches open, the circuit has a time constant 0.05 sec. With switch S_2 closed and the other two switches open, the circuit has a time constant 2 sec. With switch S_3 closed and the other two switches open, the circuit oscillates with a period T. Find T(in sec). (Take $\pi^2 = 10$)



Space for rough work

Chemistry

PART – II

SECTION – A

Straight Objective Type

This section contains **08 multiple choice questions** numbered 1 to 8. Each question has 4 choices (A), (B), (C) and (D), out of which only **ONE** is correct

1. $CH_3HC = CHCH_3 + \underset{(\text{triplet state})}{CH_2} \rightarrow P$ (cis)

(A) P is a cis isomer
(B) P is a trans isomer
(C) P is a mixture of cis and trans isomers
(D) P is acyclic compound

2. $BaTiO_3$ crystallizes in perovskite structure. This structure may be described as a cubic lattice, with barium ions occupying the corners, oxide ions at the face centres and titanium ions in one of the octahedral holes. Which of the following is correct about $BaTiO_3$ crystal lattice?

(A) Ti^{4+} must the body centred octahedral hole
(B) Ti^{4+} must occupy the edge centred octahedral hole
(C) Ti^{4+} must occupy any of the octahedral holes
(D) $1/4$ th of the octahedral holes are occupied

3. Ammonium compound which on heating does not give NH_3 is

(A) $(NH_4)_2SO_4$
(B) $(NH_4)_2CO_3$
(C) NH_4NO_2
(D) NH_4Cl

Space for rough work

4. The standard electrode potentials of some of the alkali metals are

$Li^+/Li = -3.05\,V$

$Na^+/Na = -2.71V$

$K^+/K = -2.93V$

Alkali metals react with water according to the following reaction. (M = alkali metal)

$2M + H_2O \rightarrow 2MOH + H_2$

It is found Li reacts gently with water whereas potassium reacts violently with water. Which of the following statement (s) is/ are correct regarding the above two experimental facts?

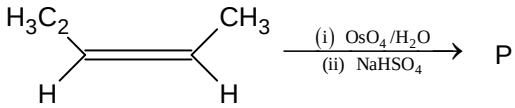
(A) ΔG values related with the kinetics of a reaction

(B) The metal having large standard oxidation potential will have lesser reactivity.

(C) K has a low melting point, and the heat of reaction is sufficient to make it melt or even vaporize. This leads to exponential increase in its surface area.

(D) The ionization potential of K is smaller than that of Li.

5.



Which of the following is true about P ?

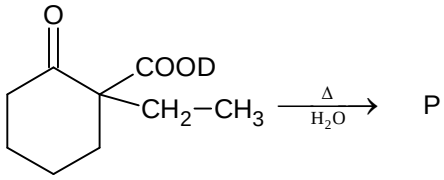
(A) P is optically inactive due to internal compensation

(B) P is optically inactive due to external compensation

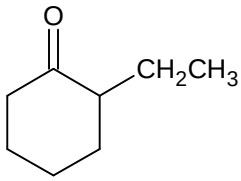
(C) P is optically inactive

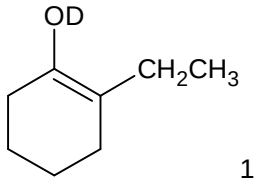
(D) P is optically inactive due the absence of chiral carbon.

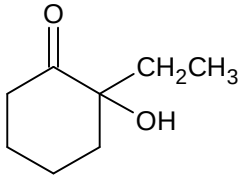
6.



Which is not possible for P is

(A) 

(B) 

(C) 

(D) None of these

1

Space for rough work

7. The salt having isoelectronic cation and anion among the following is
(A) $NaCl$ (B) KF
(C) $CaCl_2$ (D) KBr

8. Which of the following statement(s) is/ are incorrect?
(A) o – phenol sulphonic acid be converted into p – phenol sulphonic acid in the presence of H_2SO_4 if temperature is increased.
(B) Phenol is more acidic than benzyl alcohol
(C) Phenoxide ion activates the more in comparison to phenol
(D) Bromination of phenol in aqueous medium produces 2 – bromophenol and 4 - bromophenol

Multiple Correct Choice Type

This section contains **04 multiple choice questions.** Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which only **ONE OR MORE THAN ONE** is/are correct

9. Which of the following special (s) is/ are isostructural with lF_4^- ?
(A) $\left[Cu\left(NH_3\right)_4\right]^{2+}$ (B) XeF_4
(C) BF_4^- (D) $\left[Ni\left(CN\right)_4\right]^{2-}$

10. Identify the incorrect statement (s)?
(A) The ration of the observed molar volume to ideal volume is called compressibility factor.
(B) The value of van der Waals constants 'a' and 'b' is independent of the nature of gas.
(C) At high temperature and low pressure, the behaviour of any gas can be explained by $\left(P+\frac{an^2}{V^2}\right)\left(V-nb\right)=nRT$
(D) H_2 and He gases are difficult to liquefy because their Critical temperature is very small.

Space for rough work

11. In which of the following reactions, the stiochiometric ratio of H_2SO_4 to H_2 gas is the same?

(A $Zn(s)$ and H_2SO_4 (B) $Na(s)$ and H_2SO_4

(C) $Al(s)$ and H_2SO_4 (D) $Ni(s)$ and H_2SO_4

12. Which of the following compound (s) will not dissolve in an alkali solution after it has undergone reaction with Hinsberg's reagent?

(A) Diethyl amine (B) Isopropyl amine

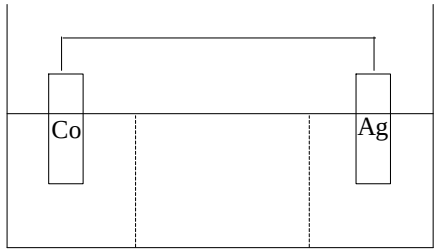
(C) N – methyl ethanamine (D) N, N – dimethyl aniline

Comprehension Type

This section contains **3 groups of questions.** Each group has 2 multiple choice question based on a paragraph. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which only **ONE** is correct

Paragraph for Questions 13 & 14

The cell given below consists of three compartments, separated by porous barriers. The first contains a cobalt electrode in $5\,L$ of $0.1\,M$ $Co(NO_3)_2$; the second contains $5\,L$ of $0.1\,M$ KNO_3 the third contains a silver electrode in $5\,L$ of $0.1\,M$ $AgNO_3$ Assume that the current within in the cell Is carried equally by the positive and negative ions. If 'there is a passage of 0.1 mol of electrons, answer the following (19, 20 and 21) questions.



$E^o_{Co^{2+}/Co} = -0.28V$

$E^o_{Ag^+/Ag} = 0.8V$

13. The concentrations of Co^{2+} and NO_3^- in the left hand side .compartment are

(A) $0.105\,M$ and $0.210\,M$ (B) $0.08\,M$ and $0.16\,M$

(C) $0.1\,M$ and $0.2\,M$ (D) $0.05\,M$ and $0.1\,M$

14. The concentrations of Ag^+ and NO_3^- In the right hand side compartment are.

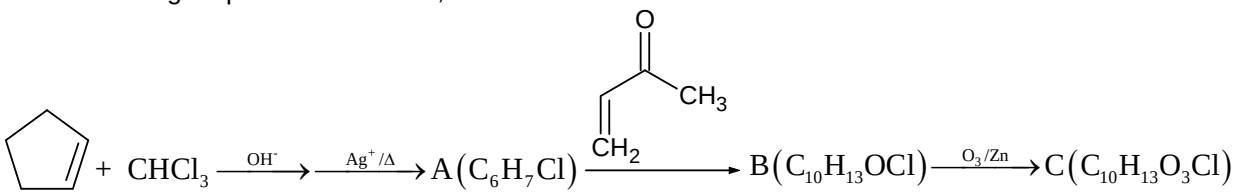
(A) $0.08\,M$ and $0.09\,M$ (B) $0.75\,M$ and $0.08\,M$

(C) $0.01\,M$ and $0.09\,M$ (D) $0.75\,M$ and $0.09\,M$

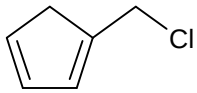
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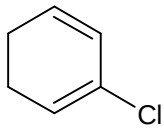
Paragraph for Questions 15 & 16

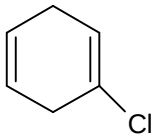
In the following sequence of reaction,



15. Compound A is

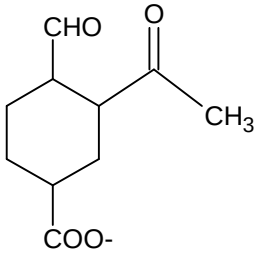
(A) 

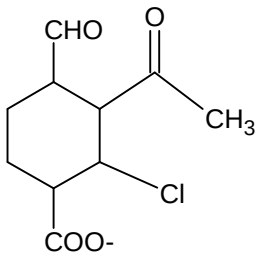
(B) 

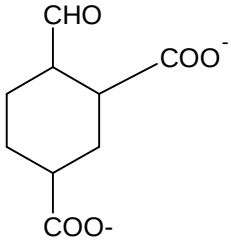
(C) 

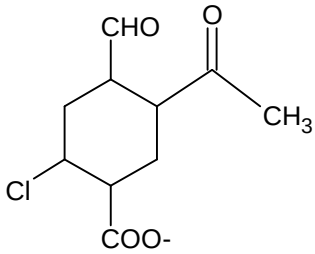
(D) None of these

16. Compound C when treated with NaOH/ X_2 gives

(A) 

(B) 

(C) 

(D) 

Space for rough work

Paragraph for Questions 17 & 18

Potassium dichromate is used as lab reagent and dichromate ion can be used as a primary standard in volumetric analysis.

$$\frac{1}{2}Cr_2O_7^{2-} + 7H^+ + 3e \rightleftarrows Cr^{3+} + 3\frac{1}{2}H_2O; E° = 1.33\,V$$

When H_2O_2 is added to an acidified solution of dichromate, a complicated reaction occurs. The formation of product depends on the pH value and concentration of dichromate ions. A deep-blue violet peroxo compound X is formed which decomposes rapidly in aqueous solution. This compound can be extracted in ether, where it reacts with pyridine fanning the adduct $PyCro(O_2)_2$.

In alkaline solution $K_2Cr_2O_7$ with 30% H_2O_2, a read – brown compound $Y(K_3CrO_8)$ is formed. In alkaline solution the dark red – brown compound $Z[(NH_3)_3CrO_4]$ is formed. In alkaline solution dichromate ion changes to chromate ion which is also used as reagent in quantitative analysis of cations.

17. The shape of compound (Z) is
(A) Butterfly shape
(B) Pentagonal pyramidal
(C) Pentagonal bipyramidal
(D) Tetrahedral

18. What is the oxidation state of Cr In red-brown compound Y?
(A) + 6
(B) + 4
(C) + 5
(D) + 13

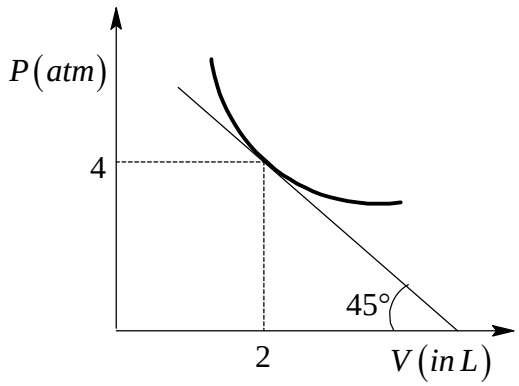
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SECTION – C

(One Integer Value Correct Type)

This section contains **5 questions**. Each question, when worked out will result in **one integer** from 0 to 9 (both inclusive).

1. The molar heat capacity of the ideal monoatomic gas undergoing the process shown in the figure is 'x'. The value of '2x' will be

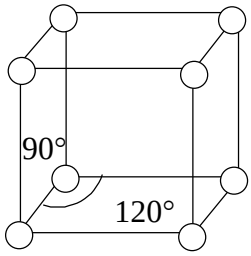


2. How many statement(s) are correct?

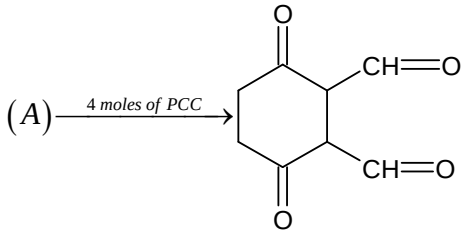
(a) if the critical micelle concentration (CMC) of a soap is $10^{-3}M$, then $10^{-4}M$ solution of this soap will have colloidal nature.
(b) charge on the colloidal particles may be determined with the help of dialysis
(c) peptization is the process of conversion of a colloidal sol into precipitate by adding some electrolyte
(d) when some oil-soluble dye is added in the water in oil emulsion, the back ground becomes coloured
(e) For the coagulation of a particular colloidal solution, the order of coagulation power of some electrolytes is in the order: $Na_3PO_4 > BaSO_4 > AlCl_3$, in electro-osmosis, the dispersion medium move towards the anode
(f) diffused layer around colloidal particles in the electric double layer mostly contains negatively charged particles.(colloid is of 'e' part)
(g) it may be basic dye such as methylene blue (colloid is of 'e' part)
(h) it may be metal sol (colloid is of 'e' part)

Space for rough work

3. Ice crystallizes is an hexagonal lattice. At the low temperatures at which the structure was determined, the lattice constants were $a = 4.53\mathring{A}$ and $b = 7.41\mathring{A}$ as shown in the figure. How many H molecules are contained in a unit cell? (Given that density of ice $= 0.92\, g / cm^3$)?



4. A weak acid type indicator was found to be 60% dissociated at pH = 9.18. If the degree of dissociation at pH = 9.0 is 'α'? Then the value of '10α' will be $(\log 2 = 0.3, \log 3 = 0.48)$

5. 

Maximum number of moles of Ac_2O consumed by reactant (A) is?

Space for rough work

Mathematics PART – III

SECTION – A

Straight Objective Type

This section contains **08 multiple choice questions** numbered 1 to 8. Each question has 4 choices (A), (B), (C) and (D), out of which only **ONE** is correct

1. Conditions so that equation $(a_1x^2+b_1x+c_1)(a_2x^2+b_2x+c_2)=0$, $(a_1a_2>0)$ have fours real roots if
(i) $c_1c_2>0$ (ii) $a_1c_2<0$ (iii) $a_2c_1<0$
(A) only (i) is sufficient
(B) either of two statements are sufficient
(C) either of two statements are necessary and sufficient
(D) all the three statements are sufficient

2. In a ΔABC, the incircle touches the sides BC, CA and AB at D, E, F respectively. If radius of incircle is 4 units and BD, CE and AF are in A.P. with common difference 1, then area of ΔABC is
(A) 84 sq. units (B) 42 sq. units
(C) 168 sq. units (D) data insufficient

3. Let A be the area between co-ordinate axis, $y^2=x-1$, $x^2=y-1$ and the line which makes the shortest distance between two parabolas and A′ be the area between $x=0$, $x^2=y-1$, $x=y$ and the shortest distance between $y^2=x-1$ and $x^2=y-1$, then
(A) A=A′ (B) $A=(A')^{1/2}$
(C) $A=2A'$ (D) can't say anything

4. a,b,c are in G.P. and the equation $(a+di)x^2+2(b+ei)x+(c+if)=0$ have one real root then $\frac{a}{d},\frac{b}{e},\frac{c}{f}$ are in, $(a,b,c,d,e,f\in R)$
(A) AP (B) GP
(C) HP (D) none of these

Space for rough work

5. $\int \sec^{p/3} x \operatorname{cosec}^{q/3} x dx = (p,q \in I^+ \text{ and } p+q=12)$

(A) $\dfrac{\tan^{\frac{-p}{3}+1} x}{\frac{-p}{3}+1} + c$

(B) $\dfrac{\tan^{\frac{-p}{3}+1} x}{\frac{-p}{3}+1} + c + \dfrac{\tan^{\frac{-p}{3}+3} x}{\frac{-p}{3}+3} + c$

(C) $\dfrac{\tan^{\frac{-p}{3}+3} x}{\frac{-p}{3}+3} + c$

(D) none of these

6. A square ABCD of a diagonal 2a is folded along the diagonal AC, so that the planes DAC, BAC are at right angle. The shortest distance between DC and AB is

(A) $\sqrt{\frac{2}{3}}a$

(B) $\dfrac{2a}{\sqrt{3}}$

(C) $\sqrt{\dfrac{2a}{3}}$

(D) none of these

7. If $P = \int_0^\infty \frac{x^2 dx}{1+x^4}, Q = \int_0^\infty \frac{x dx}{1+x^4}$ and $R = \int_0^\infty \frac{dx}{1+x^4}$, then which of the following is false.

(A) $Q = \dfrac{\pi}{4}$

(B) P=R

(C) $P - \sqrt{2}Q + R = \dfrac{\pi}{2\sqrt{2}}$

(D) none of these

8. If $\int_{\pi/2}^{x} \sqrt{3\text{-}2\sin^2 t} dt + \int_0^y \cos t dt = 0$, then $\left(\dfrac{dy}{dx}\right)$ at $x = \pi$ and $y=\pi$ is

(A) $\sqrt{3}$

(B) $-\sqrt{2}$

(C) $-\sqrt{3}$

(D) none of these

Space for rough work

Multiple Correct Choice Type

This section contains **04 multiple choice questions.** Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which only **ONE OR MORE THAN ONE** is/are correct

9. Let z_1, z_2, z_3 be the coordinate of vertices *A*, *B*, *C* of the triangle if $W_1 = z_1 - z_2$, $W_2 = z_3 - z_2$ and $\angle A = 90°$, then

(A) $R_e\left(\frac{W_2}{W_1}\right) = 1$ (B) $R_e(W_1\overline{W_2}) = |W_1|^2$

(C) $R_e(W_1\overline{W_2}) = |W_2|^2$ (D) $R_e\left(\frac{W_1}{W_2}\right) = 1$

10. Let $(1 + x + x^2)^n = a_0 + a_1x + a_2x^2 \ldots\ldots a_{2n}x^{2n}$, where *n* is odd integer

$S_1 = a_0 + a_4 + a_8 + \ldots\ldots\ldots$

$S_2 = a_1 + a_5 + a_9 + \ldots\ldots\ldots$

$S_3 = a_2 + a_6 + a_{10} + \ldots\ldots\ldots$

$S_4 = a_3 + a_7 + a_{11} + \ldots\ldots\ldots$

(A) $S_1 = S_3$ (B) $S_2 = S_4$

(C) $S_2 + S_4 = 0$ (D) $S_1 = S_2 = S_3$ or $S_1 = S_3 = S_4$

11. If $f(x) = ax^2 + bx + c = 0$ have real roots and its coefficients are odd the integers then

(A) $f(x) = 0$ always have irrational roots

(B) $\left|f\left(\frac{p}{q}\right)\right| \geq \frac{1}{q^2}$ where $p, q \in I$

(C) If $a.c. = 1$, then equation must have atleast are root α such that $[\alpha] = -1$, where [.] is GIF

(D) Equation have rational roots

Space for rough work

12. The curve $y = (|x|-1)\operatorname{sgn}(x-1)$ divides $\frac{9x^2}{64}+\frac{4}{25}y^2=\frac{1}{\pi}$ in two parts having area A_1 and A_2 (where $A_1 < A_2$), then

(A) $\frac{A_1}{A_2}=\frac{7}{13}$ (B) $\frac{A_1}{A_2}=\frac{3}{7}$

(C) $A_1=\frac{7}{3}$ (D) $A_2=\frac{13}{7}$

Comprehension Type

This section contains **3 groups of questions.** Each group has 2 multiple choice question based on a paragraph. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which only **ONE** is correct

Paragraph for Questions 13 & 14

If a circle C_o, with radius 1 unit touches both the axes and as well as line (L_1) through $P(0,4)$, cut the x-axis at $(x_1, 0)$. Again a circle C_1 is drawn touching x-axis, line L_1 and another line L_2 through point *P*. L_2 intersects x-axis at $(x_2, 0)$ and this process is repeated *n* times.

13. The centre of circle C_2 is

(A) $(3, 1)$ (B) $\left(\frac{15}{2}, 1\right)$

(C) $\left(\frac{31}{4}, 1\right)$ (D) none of these

14. The $\lim_{n\to\infty}\frac{x_n}{2^n}$ is

(A) 4 (B) 0
(C) 1 (D) 2

Space for rough work

Paragraph for Questions 15 & 16

f(x), g(x), h(x) all are continuous and differentiable functions in [a, b] also a<c<b and f(a)=g(a)=h(a). Point of intersection of the tangent at x=c with chord joining x=a & x=b is on the left of c in y=f(x) and on the right in y=h(x). And tangent at x=c is parallel to the chord in case of y=g(x).

15. If f(b) = g(b) = h(b) then

(A) $f'(c) = g'(c) = h'(c)$ (B) $f'(c) > g'(c) > h'(c)$

(C) $f'(c) < g'(c) < h'(c)$ (D) none of these

16. If $c = \frac{a+b}{2}$ for each b, then

(A) $g(x)=Ax^2+Bx+C$ (B) $g(x)=\log x$

(C) $g(x)=\sin x$ (D) $g(x)=e^x$

Paragraph for Questions 17 & 18

Let f: R $\to$ R be a function satisfying condition $f(x + y^3) = f(x) + (f(y))^3 \ \forall\ x, y \in R$ and $f'(0) = k$.

17. If |k| > 0, then $\lim_{x\to 0}\left[\frac{\tan x}{f(x)}\right]$ (where [.] denotes the greatest integer function) is

(A) 0 (B) 2

(C) – 2 (D) none of these

18. If |k| < 0 then value of $f\left(\frac{\sqrt{3}-1}{\sqrt{2}+\sqrt{5}}\right)$ is/are

(A) $-\left(\frac{\sqrt{3}-1}{\sqrt{2}+\sqrt{5}}\right)$ (B) 0

(C) $\left(\frac{\sqrt{3}-1}{\sqrt{2}+\sqrt{5}}\right)$ (D) ± 1

Space for rough work

SECTION – C

(One Integer Value Correct Type)

This section contains **5 questions**. Each question, when worked out will result in **one integer** from 0 to 9 (both inclusive).

1. Let *M* be the midpoint of sides *QR* of a rectangle *PQRS* respectively. A line through *M*, intersect *RS* and *PS* extended at *T* and *L* respectively such that $PL = \frac{4}{3}PS$. Find the value of $20\left(\frac{\text{area of } \Delta\, MRT}{\text{area of } \square PQRS}\right)$.

2. If the equations $x^3 + 5x^2 + px + q = 0$ and $x^3 + 7x^2 + px + r = 0$ have two roots in common, then find the difference of two uncommon roots.

3. When 23^{23} is divided by 53, the remainder is *r*. Find $\frac{r}{10}$

4. The number of pairs (*x*, *y*) satisfying the equation $\sin x + \sin y = \sin(x + y)$ and $|x| + |y| = 1$ is

5. The value of $33\left\{\sum_{n=1}^{16}\left\{\frac{4n^4}{4n^2-1}\right\}\right\}$ is ……. (where {.} represents fractional part function)

Space for rough work